BJGP OPEN

Learning to navigate uncertainty in primary care: a scoping literature review

Gardner, Nick P; Gormley, Gerard J; Kearney, Grainne P

DOI: https://doi.org/10.3399/BJGPO.2023.0191

To access the most recent version of this article, please click the DOI URL in the line above.

Received 21 September 2023
Revised 28 November 2023
Accepted 12 December 2023

© 2023 The Author(s). This is an Open Access article distributed under the terms of the Creative Commons Attribution 4.0 License (http://creativecommons.org/licenses/by/4.0/). Published by BJGP Open. For editorial process and policies, see: https://bjgpopen.org/authors/bjgp-open-editorial-process-and-policies

When citing this article please include the DOI provided above.

Author Accepted Manuscript
This is an 'author accepted manuscript': a manuscript that has been accepted for publication in BJGP Open, but which has not yet undergone subediting, typesetting, or correction. Errors discovered and corrected during this process may materially alter the content of this manuscript, and the latest published version (the Version of Record) should be used in preference to any preceding versions

Learning to navigate uncertainty in primary care: a scoping literature review.

Nick P. Gardner[1]; Gerard J. Gormley[1]; Grainne P. Kearney[1].

[1] Centre for Medical Education, Queen's University Belfast, Whitla Medical Building, 97 Lisburn Road, Belfast BT9 7BL, Northern Ireland

Corresponding author:

Dr Nick Gardner, n.gardner@qub.ac.uk

Keywords:

Uncertainty; medical education; general practice; medical student; scoping review

Wordcount: 2658

Accepted Manuscript - BJGP Open - BJGPO.2023.0191

Abstract

Background: Clinical practice occurs in the context of uncertainty. Primary care is a clinical environment that accepts and works with uncertainty differently from secondary care. Recent literature reviews have contributed to understanding how clinical uncertainty is taught in educational settings and navigated in secondary care, and, to a lesser extent, by experienced GPs. We don't know how medical students and doctors in training learn to navigate uncertainty in primary care.

Aim: In this scoping review, we asked the question: What is known about primary care as an opportunity for learning to navigate uncertainty?

Design and setting: Scoping review of articles written in English.

Methods: Using a scoping review methodology, Embase, Medline and Web of Science databases were searched, with additional articles obtained through citation searching. Studies were included in this review if they (a) were based within populations of medical students and/or doctors in training, and (b) considered clinical uncertainty or ambiguity in primary care, or a simulated primary care, setting. Study findings were analysed thematically.

Results: Thirty-six studies were included from which three major themes were developed: uncertainty contributes to professional identity formation (PIF), adaptive responses, and maladaptive behaviours. Relational and social factors that influence PIF were identified. Adaptive responses included adjusting epistemic expectations, and shared decision-making.

Conclusion: Educators can play a key role in helping learners navigate uncertainty through socialisation, discussing primary care epistemology, recognising maladaptive behaviours and fostering a culture of constructive responses to uncertainty.

Accepted Manuscript - BJGP Open - BJGPO.2023.0191

How this fits in

Learning to navigate clinical uncertainty is an essential aspect of medical education. This is the first scoping review to consider what is known about Primary Care as an opportunity for learning to navigate uncertainty during undergraduate and postgraduate training. The findings suggest that clinical uncertainty influences medical students' and GP trainees' professional identity formation, with both adaptive and maladaptive responses identified. It is important that GP educators are aware of these outcomes when considering the GP training environment and the need for formal teaching about navigating uncertainty.

Introduction

In clinical practice, physicians make decisions in the context of incomplete information, imperfect cognitive processing, and the potential for unpredictable patient responses. Thus, clinical encounters are always viewed through a lens of uncertainty, with varying degrees of translucency. The ability to navigate uncertainty is an indispensable component of clinical practice and is associated with a spectrum of cognitive, emotional and behavioural responses.[1]

Uncertainty has been defined as the "dynamic, subjective perception of not knowing what to think, feel, or do".[2] It has two requirements: the absence of understanding and the presence of conscious awareness.[3] The extent to which a physician can tolerate uncertainty has recognised influences, including choice of specialisation[4] and psychological well-being.[5] Patient care is also affected, through timeliness of diagnoses, use of healthcare resources, and prescribing.[6] General Practitioners (GPs) with lower tolerance of uncertainty are more likely to refer patients to secondary care colleagues.[7] These are not benign activities, with potential adverse physical and psychological outcomes for patients and impact on an already strained healthcare systems.

Exposure to uncertainty in medical training is necessary for cultivating the professional virtues drawn upon when navigating uncertainty.[8] Individuals are socialised into the culture of their clinical environments, and this includes the extent to which uncertainty is accepted.[9] Park[10] describes primary care as a "cultural setting regularly encountering and using uncertainty within consultations". Johnston[11] explains that "uncertainty and complexity are defining features of the primary care paradigm". However, recent literature reviews about navigating uncertainty in training have focussed on secondary care settings[12, 13] or undergraduate medical educational interventions.[14] Alam et al.[15] found that experienced GPs relied on established relationships and heuristics, however medical students and doctors in training are rarely afforded time in practice to develop these techniques. We propose that individuals learning to practice medicine in primary care may navigate clinical uncertainties differently to experienced clinicians. However, we don't know what is known about primary care as an opportunity for learning to navigate uncertainty in medical training.

Methods

This scoping review was based on the six-step methodological framework developed by Arksey and O'Malley[16] and advanced by Levac et al[17]. The review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) extension for scoping reviews.[18] Ethical approval was obtained for stage six of this scoping review (MHLS 23_59). The researchers in this study included NG, a GP and postgraduate research student, GK, a GP and clinical senior lecturer, and GG, a GP and professor of simulation and clinical skills.

Accepted Manuscript - BJGP Open - BJGPO.2025.0191

Identifying the research question

This scoping review sought to answer the question: what is known about primary care as an opportunity for learning to navigate uncertainty? In doing so, its aims were to (a) identify the extent, range and nature of the literature available about clinical uncertainty in primary care experienced by medical students and doctors in training and (b) develop themes which may inform primary care medical educators and influence future research in this area.

Identifying relevant studies

Studies were included in this review if they (a) were based within populations of medical students and/or doctors in training, and (b) considered clinical uncertainty or ambiguity in primary care, or a simulated primary care, setting. Studies based in secondary care, or solely involving senior physicians were excluded, as were non-English full-text studies.

Search criteria were formulated for three databases: Embase, Medline and Web of Science, from database inception to present day. Table 1 contains the search terms used. Support was provided by a subject librarian. Although ambiguity's status as a synonym for uncertainty has generally been adjusted to that of a source or stimulus, it was included in this search as it has been used interchangeably with uncertainty in scientific literature. Search results from each database were imported to the bibliographic management system Covidence.[19]

Study selection

Inclusion criteria and search terms were developed in an iterative way during regular research team meetings. The initial screening of studies was conducted by NG. 62 studies met criteria for full text review, which was conducted by NG and GG independently, with GK arbitrating on studies where NG's and GG's decisions differed. All three reviewers were in agreement about the 36 studies included. The reasons for exclusion at this stage at outlined in Figure 1. Citation searching was conducted during the full text review stage which yielded two additional studies.

Accepted Manuscript - BJGP Open - BJGPO.2023.0191

Figure 1 PRISMA flow diagram[20]

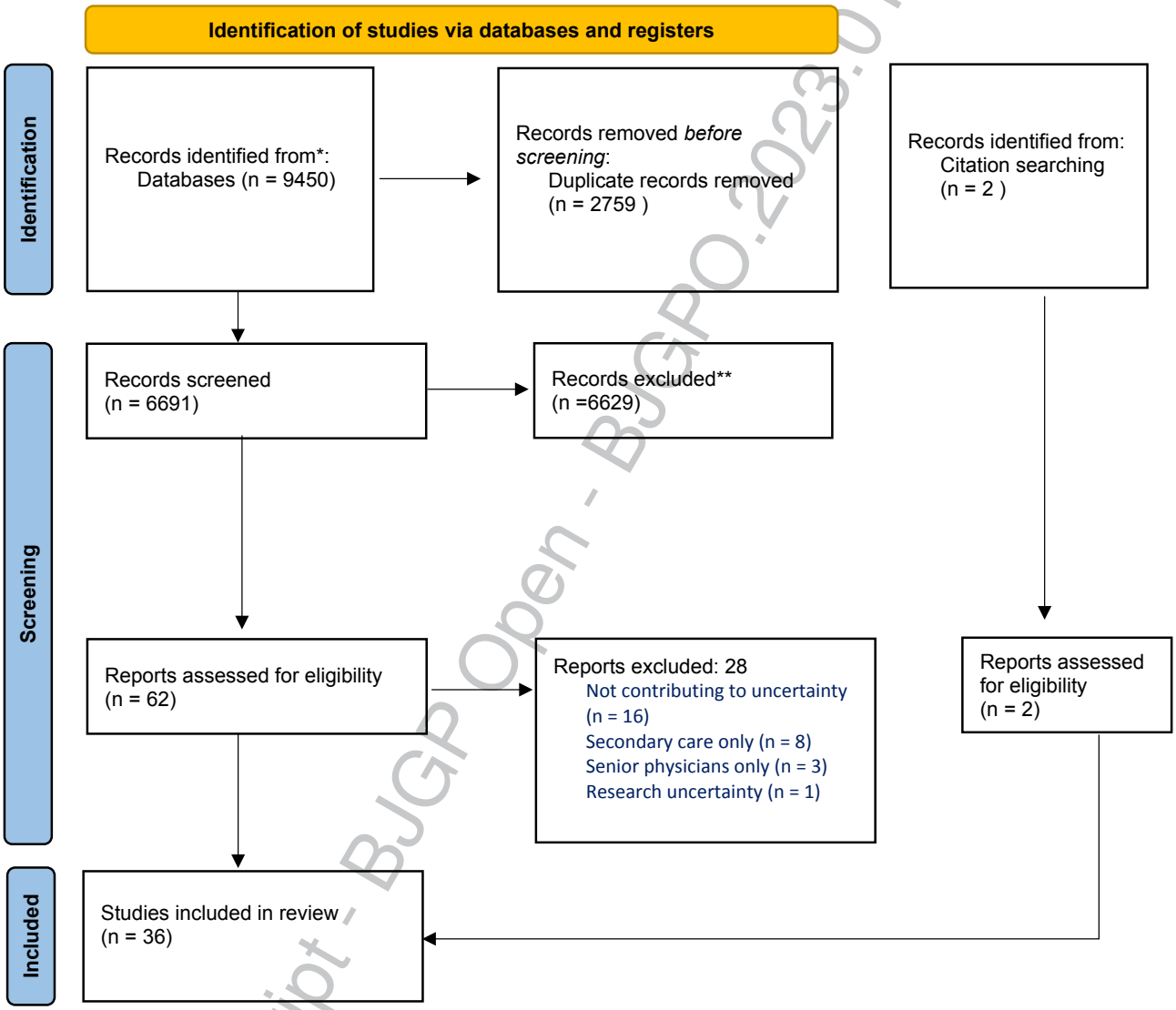

Table 1: search terms

Primary care	Uncertainty	Education
• primary care/primary medical care/primary healthcare • general pract* • family medic*	• uncertain* • ambig*	• education • curricul* • training • trainee • student

Accepted manuscript - BJGP Open - BJGPO.2023.0191

Charting the data

A data charting excel spreadsheet was produced, based on criteria recommended by the JBI Scoping Review Network.[21] Study findings were analysed thematically, as indicated in Table S1. This was an iterative process, with themes initially developed from ten randomly selected papers with further refinements made throughout. The remaining papers were coded with these themes by one researcher (NG) and discussed as a research team during regular meetings. The team worked reflexively[22]: shaping themes and discussing issues, ensuring multiple perspectives on the data and rigorous analysis.

Results

Search results

Broad keywords used in database searches yielded an initial 6691 studies after duplicates were removed. Once inclusion and exclusion criteria were applied this resulted in 62 studies for full-text review. This yielded 34 studies; two more from reviewing citations brought the total to 36. Study characteristics are shown in Table 2.

Table 2: Study characteristics

Populations	Medical students (12/36) and postgraduate primary care trainees (24/36)
Publication date	Majority after year 2000 (34/36)
Journal affiliations	Majority were family medicine (20) or medical education (10)
Geographical locations	UK (11), Australia (8), USA (5), Canada (3), Ireland, Germany and Norway (2 each), Sweden, France and Finland (1 each).
Study methodologies	Quantitative (15), qualitative (14), mixed methods (3), opinion pieces (3), and one systematic review (of 20 studies).

Results of thematic analysis

Uncertainty contributes to professional identity formation

Eight[10, 23, 24, 25, 26, 27, 28, 29] studies found that clinical uncertainty contributes to professional identity formation (PIF) in primary care. Doctors training in primary care recognised their "specific role in the broader context of medicine, that when situated with specialists' narrower expertise"[24] allows them to accept uncertainty. Clinical uncertainty, and its impact on trainees' "being able to sleep at night", was found to expedite PIF development as they sought to "position themselves on a cavalier-to-overcautious spectrum".[29] GP trainees reflected on their professional identity, developing an appreciation of belonging to the wider GP community.[25] Longitudinal undergraduate primary care placements, where uncertainty contributed to transformed thinking and newfound confidence in the medical student role, were described as "ideal pedagogic spaces to negotiate the thresholds involved in professional identity formation"[27].

Professional identity formation is a "dynamic process achieved through socialization"[30], and ten [24, 28, 29, 31, 32, 33, 34, 35, 36, 37] studies specifically referred to the social role that primary care plays in medical students and GP trainees learning about navigating uncertainty. This relationship-focussed strategy[12] encompasses the role of social resources[38] pooled from the wider primary care team. GP trainees reported becoming more comfortable with uncertainty by observing senior colleagues' admissions of uncertainty, appreciating the realisation that "no-one's thinking they know it all".[29]

Accepted Manuscript - BJGP Open - BJGPO.2023.0191

This process of socialisation: "communicative behaviours that contributed to a culture comfort with uncertainty", was not information-seeking; it's purpose was "the affective re-appraisal of the role of uncertainty".[24] For medical students, "identification with more advanced members of 'the tribe'" helped to convert uncertainty into resilience.[28]

Adaptive responses

Nine studies[27, 29, 34, 35, 36, 39, 40, 41, 42] contributed to understanding the influence that time spent in primary care has on navigating clinical uncertainty. A unanimous finding among these studies was that experience has a positive moderating effect on the discomfort initially associated with navigating uncertainty.

Quantitative analysis[39] of 594 GP trainees showed that anxiety due to uncertainty, concern about a bad outcome, and reluctance to disclose diagnostic or treatment uncertainty to patients improves over the duration of training. DeForge and Sobal[41] found this was independent of the trainees' age. Medical students undertaking extended placements in primary care developed growing comfort with uncertainty. Over the course of a year their perception of uncertainty changed "from threatening to challenging to exciting".[27]

Whilst experience was independently associated with greater tolerance of uncertainty, additional adaptations to exposure to uncertainty were identified: shared decision-making and adjusting epistemic expectations. Shared decision-making (SDM) was a finding in nine studies.[10, 26, 32, 43, 44, 45, 46, 47, 48]Often viewed as a "middle ground", SDM is a consultation model found midway between paternalism and informed choice with an emphasis on involving patients in decision-making,[45] and can be considered a "functional" adaptation to clinical uncertainty in GP trainees, helping both the doctor and the patient.[32]

Guenter et al.[43] encourage GP trainees "to *think out loud* with the patient about the analytical process" when navigating uncertainty. Elwyn et al.[45] found that GP trainees recognised that SDM "unburdens the doctor." Several studies in the undergraduate population highlighted the importance of GP tutors verbalising their decision-making processes with medical students. Parekh et al.[48] called for GPs tutors to make the "implicit explicit" so that medical students can learn to understand "the cognitive processes they are working through when seeing patients with… medical uncertainty". Park[10] explained that doing so legitimises exploration of uncertainty.

Eleven studies[10, 25, 27, 28, 29, 33, 36, 43, 46, 49, 50] also identified adjusting epistemic expectations as an adaptation to uncertainty. Han et al.[12] define this as "acknowledging the impossibility of perfect medical knowledge and thereby relinquishing the quest for certainty." The "uncertain world of general practice" creates an environment in which medical knowledge in itself was not an indicator of quality; as a GP trainee explained: "(you) realise that a lot of these rules and things that we are taught are very black and white, and that's not true in the context of the person's life and circumstances…"[33] Trainees became comfortable with uncertainty by developing an understanding of the consultation process, by "'readjusting (their) perspective' to solving problems in general practice over a number of consultations".[29] This also involved adjusting ideas about the role of a competent GP: "registrars had to leave behind the value that 'good' medicine always involved a clear diagnosis and a cure".[36] Medical students considered it a revelation that uncertainty can be accepted as a legitimate professional strategy, [10] and gained confidence by recognising that "it's not always about having a correct answer, its more the process".[27]

Accepted Manuscript - BJGP Open - BJGPO.2023.0191

Maladaptive behaviours

Ten studies[23, 32, 35, 39, 43, 45, 47, 51, 52, 53] identified the presence of maladaptive behaviours in medical students and GP trainees. These behaviours took two forms: premature closure of diagnostic reasoning associated with actions (admitting, referring, investigating, or prescribing) motivated at imposing certainty too early, and reluctance to disclose uncertainty. [28, 32, 40, 41]

Several studies described GP trainees taking cognitive and behavioural shortcuts in the presence of clinical uncertainty, described as a "flailing attempt to impose a higher level of certainty on a situation than that situation is ready for".[43] Danczak and Lea[32] noted that similar management strategies "seemed to replace analysis with less cognitively demanding, intuitive thinking, leading to action, ending the stress of uncertainty". Trainees admitted that they used referrals as "a way of absolving themselves of uncertainty",[52] without considering the negative consequences this could have. Carr and Gormley[47] also identified this affective stimulus in medical students, whose decisions to investigate patients were "influenced significantly by the discomfort that they experienced whilst 'holding the risk' associated with clinical uncertainty." One study[23] showed greater reluctance to disclose uncertainty to patients was associated with burnout in GP trainees.

Consultation with stakeholders

When presented in May 2023, the themes of this study resonated with a group of 30 GP educators experienced in training medical students and doctors in training. Moreover, they agreed with the maladaptive behaviours described, particularly around excessive investigation and referral, and that socialisation improves this. GP trainers observed that primary care epistemology changes with experience, and that observing different consultation styles influences GP trainees' PIF.

Discussion

Summary

This scoping review provides an overview of the existing literature on clinical uncertainty in primary care, as studied in medical students and doctors in training. Thirty-six studies met the inclusion criteria and led to the development of three themes observed in both the undergraduate and postgraduate populations: uncertainty contributes to PIF, adaptive responses, and maladaptive behaviours.

Strengths and limitations

This scoping review is the first to review clinical uncertainty in medical students and doctors in training based in primary care. Methodological strengths include an extensive initial search strategy combined with a review of citations within included studies. However, we may not have included all relevant studies. Although a single-reviewer process was adopted for screening and data extraction, blinded, double screening of full-text reviews and iterative review meetings discussing data charting and thematic analysis added rigor to the methodology. Sharing preliminary findings with stakeholders helped to validate the findings, although consultation was limited to educators. Several included studies did not contribute to any of the themes, and two papers were only available as abstracts.[34, 54]

Accepted Manuscript - BJGP Open - BJGPO.2023.0191

Comparison with existing literature

Cruess et al.[30] describe PIF as influenced by three domains: individual (personal characteristics), relational (the influence of a mentor) and collective (the impact of social groups). Findings in this review tended to concentrate on the latter two domains. Medical students and doctors in training valued forming professional relationships in the pursuit of navigating uncertainty, which influenced their professional identity. This was not done exclusively to procure a source of knowledge. Rossignac-Millon and Higgins[55] explain that "humans are truth-cartographers searching for epistemic companions with whom to map out the bounds of reality". In the same way, medical students and doctors training in primary care form relationships that enable them to better understand the cultural acceptance of uncertainty, and through doing so are socialised into a shared way of thinking and behaving.

Recent frameworks for understanding uncertainty and its tolerance have highlighted moderators of the perception of, and a spectrum of reactions (cognitive, emotional and behavioural) to, uncertainty.[1, 2] This literature review correlates with these frameworks, and identifies responses in each of these reaction domains which are both adaptive and maladaptive. This review highlights PIF, partly through socialisation, as an important moderator of how uncertainty is perceived. The results also show an overlap of these 'response' domains, for example the cognitive response of adapting epistemic expectations was inseparable from the emotional responses involved: epistemic adjustments influenced the affective appraisal of uncertainty.

The findings of this review also align with a recent study by Ilgen et al.[56] Studying comfort with uncertainty, the authors advocate the need for trainees to adapt their epistemic expectations, calling for educators to "disabuse their trainees of the notion that clinical knowledge exists in a binary format of 'knowing' and 'not knowing'". They also identify similar maladaptations in response to uncertainty: certainty as a precondition to action, premature diagnostic closure and over-testing.

Several studies highlighted areas in which medical students and doctors in training were felt to be ill-prepared for managing clinical uncertainty. These included perceptions about the need to "'fix' the patient", [43, 52] and misconceptions about SDM.[45] As a result, medical students and doctors in training risk experiencing an educational dichotomy between what they are taught and what they experience in practice.[57]

Implications for research and practice

The results inform several recommendations for undergraduate and postgraduate medical educators. Involving students and trainees in team discussions could provide valuable educational opportunities. Explaining differences in primary and secondary care epistemologies may help in the early navigation of uncertainty. GP educators might benefit from awareness of recognised maladaptive behaviours, and by deliberately fostering a culture of functional responses to uncertainty. Disclosing the degree of uncertainty in the diagnosis should be recommended as an essential aspect of safety-netting,[58] the benefits of shared decision-making within the GP consultation should be emphasised.

It is important to consider whether these recommendations reflect the formal teaching that medical students and GP trainees are receiving. Clinicians and their patients suffer when uncertainty is inexpertly navigated. It is therefore necessary to enquire whether this is reflected in formal undergraduate and postgraduate curricula, considered to be learned best 'in-situ', or simply an overlooked aspect of medical education? This scoping review highlights the valuable learning opportunities presented in the everyday exposure to uncertainty in primary care. However, in light

Accepted Manuscript - BJGP Open - BJGPO.2023.0191

of the potential for maladaptive behaviours, is a greater emphasis on evidence-based formal training needed for both GP educators and their learners?

This review also identifies the potential influence that primary care can have on a learner's epistemological beliefs in the undergraduate and postgraduate populations. Eastwood et al.[59] report that studies relating to uncertainty have tended to focus on reactions to uncertainty, rather than the nature of epistemic cognition. Future research looking at how uncertainty in primary care influences the nature of epistemic cognition may help inform aspects of medical education programmes.

Conclusion

This scoping review offers an overview of how medical students and GP trainees navigate clinical uncertainty within primary care. Thematic analysis identifies that uncertainty can be formative in developing professional identity, and that response to it can be adaptive and maladaptive. We have provided concepts that have relevance in helping to shape training in primary care: where uncertainty is a "constant companion"[60], and the only certainty is that there will always be uncertainty.

Acknowledgements

We would like to thank Dr Michele Stone for allowing us permission to carry out our consultation stage, and the contributions made by the GP Educators therein.

Funding

Not applicable. This research was conducted as part of an unfunded Doctor of Medicine degree being undertaken at Queen's University, Belfast by NG.

Conflicts of interest

None.

Ethics statement

No human subjects were involved. Ethical approval for the consultation stage of this research was provided by the Faculty of Medicine, Health & Life Sciences (MHLS) Research Ethics Committee, Queen's University, Belfast.

Author contributions

All authors were involved from the conception and design of the project through multiple team meetings. NG wrote the main body of the paper with substantial amendments made by GG and GK. All authors reviewed and approved the final paper before submission. We have all agreed to be accountable for all aspects of the paper.

Accepted Manuscript - BJGP Open - BJGPO.2023.0191

References

1. Hillen MA, Gutheil CM, Strout TD, et al. Tolerance of uncertainty: Conceptual analysis, integrative model, and implications for healthcare. Soc Sci Med. 2017;180:62-75.
2. Lee C, Hall K, Anakin M, Pinnock R. Towards a new understanding of uncertainty in medical education. J Eval Clin Pract. 2021;27(5):1194-204.
3. Han PKJ. Uncertainty in medicine. A Framework for Tolerance. New York: Oxford University press; 2021.
4. Borracci RA, Ciambrone G, Arribalzaga EB. Tolerance for Uncertainty, Personality Traits and Specialty Choice Among Medical Students. J Surg Educ. 2021;78(6):1885-95.
5. Hancock J, Mattick K. Tolerance of ambiguity and psychological well-being in medical training: A systematic review. Med Educ. 2020;54(2):125-37.
6. Bhise V, Rajan SS, Sittig DF, et al. Defining and Measuring Diagnostic Uncertainty in Medicine: A Systematic Review. J Gen Intern Med. 2018;33(1):103-15.
7. Forrest CB, Nutting PA, von Schrader S, et al. Primary Care Physician Specialty Referral Decision Making: Patient, Physician, and Health Care System Determinants. Med Decis Making. 2006;26(1):76-85.
8. West AF, West RR. Clinical decision-making: coping with uncertainty. Postgrad Med J. 2002;78(920):319-21.
9. Johnston JL, Bennett D. Lost in translation? Paradigm conflict at the primary-secondary care interface. Med Educ. 2019;53(1):56-63.
10. Park S. Embracing Uncertainty within Medical Education.Giardino AP, and Giardino, E, R. Medical Education: Global Perspectives, Challenges and Future Directions. New York: Nova Biomedical; 2013.
11. Johnston JL. Conflict, Culture and Identity in GP Training. Singapore: Palgrave MacMillan; 2022.
12. Han PKJ, Strout TD, Gutheil C, et al. How Physicians Manage Medical Uncertainty: A Qualitative Study and Conceptual Taxonomy. Med Decis Making. 2021;41(3):275-91.
13. Stephens GC, Sarkar M, Lazarus MD. Medical Student Experiences of Uncertainty Tolerance Moderators: A Longitudinal Qualitative Study. Front Med. 2022;9.
14. Patel P, Hancock J, Rogers M, Pollard SR. Improving uncertainty tolerance in medical students: A scoping review. Med Educ. 2022;56(12):1163-73.
15. Alam R, Cheraghi-Sohi S, Panagioti M, et al. Managing diagnostic uncertainty in primary care: a systematic critical review. BMC Fam Pract. 2017;18(1):79.
16. Arksey H, O'Malley L. Scoping studies: towards a methodological framework. Int J Soc Res Methodol. 2005;8(1):19-32.
17. Levac D, Colquhoun H, O'Brien KK. Scoping studies: advancing the methodology. Implementation Sci. 2010;5(1):69.

Accepted Manuscript - BJGP Open - BJGPO.2023.0191

18. Tricco AC, Lillie E, Zarin W, et al. PRISMA Extension for Scoping Reviews (PRISMA-ScR): Checklist and Explanation. Ann Intern Med. 2018;169(7):467-73.
19. Covidence. systematic review software Melbourne, Australia: Veritas Health Innovation; [Available:www.covidence.org [Accessed 9 May 2023].
20. Page MJ, McKenzie JE, Bossuyt PM, et al. Updating guidance for reporting systematic reviews: development of the PRISMA 2020 statement. J Clin Epidemiol. 2021;134:103-12.
21. Peters MDJ, Marnie C, Tricco AC, et al. Updated methodological guidance for the conduct of scoping reviews. JBI Evid Synth. 2020;18(10):2119-26.
22. Olmos-Vega FM, Stalmeijer RE, Varpio L, Kahlke R. A practical guide to reflexivity in qualitative research: AMEE Guide No. 149. Med Teach. 2023;45(3):241-51.
23. Cooke GPE, Doust JA, Steele MC. A survey of resilience, burnout, and tolerance of uncertainty in Australian general practice registrars. BMC Med Educ. 2013;13(1):2.
24. Ledford CJ, Cafferty LA, Seehusen DA. Socializing Identity Through Practice: A Mixed Methods Approach to Family Medicine Resident Perspectives on Uncertainty. Fam Med. 2015;47(7):549-53.
25. Patel R. General practice trainees' learning experiences of formative think-aloud script concordance testing. Educ Prim Care. 2022;33(4):229-36.
26. Atkinson K, Ajjawi R, Cooling N. Promoting clinical reasoning in general practice trainees: role of the clinical teacher. Clin Teach. 2011;8(3):176-80.
27. Gupta S, Howden S. Medical students' experiences of a longitudinal integrated clerkship: a threshold concepts analysis. Educ Prim Care. 2021;32(6):336-43.
28. Schei E, Knoop HS, Gismervik MN, et al. Stretching the Comfort Zone: Using Early Clinical Contact to Influence Professional Identity Formation in Medical Students. J Med Educ Curric Dev. 2019;6:2382120519843875.
29. Sturman N, Jorm C, Parker M. With a grain of salt? Supervisor credibility and other factors influencing trainee decisions to seek in-consultation assistance: a focus group study of Australian general practice trainees. BMC Fam Pract. 2020;21(1):28.
30. Cruess RL, Cruess SR, Boudreau JD, et al. A Schematic Representation of the Professional Identity Formation and Socialization of Medical Students and Residents: A Guide for Medical Educators. Acad Med. 2015;90(6):718-25.
31. Rothlind E, Fors U, Salminen H, et al. The informal curriculum of family medicine – what does it entail and how is it taught to residents? A systematic review. BMC Fam Pract. 2020;21(1):49.
32. Danczak A, Lea A. What do you do when you don't know what to do? GP associates in training (AiT) and their experiences of uncertainty. Educ Prim Care. 2014;25(6):321-6.
33. Bearman M, Dracup M, Garth B, et al. Learning to recognise what good practice looks like: how general practice trainees develop evaluative judgement. Adv Health Sci Educ Theory Pract. 2022;27(1):215-28.
34. Loftus A, Keaveney L, O'Neill M. 1770 "You'll Be Fine" (Please Don'T Die!); The Experience of Uncertainty Among Gp Trainees. Arch Dis Child. 2012;97(Suppl 2):A501-A.
35. Escourrou E, Bergeaut M, Gimenez L, et al. Evolution of Reactions to Uncertainty Among Residents During a Clinical Rotation. Fam Med. 2020;52(5):339-45.
36. Stone L. Managing the consultation with patients with medically unexplained symptoms: a grounded theory study of supervisors and registrars in general practice. BMC Fam Pract. 2014;15(1):192.
37. Ledford CJ, Seehusen DA, Chessman AW, Shokar NK. How we teach U.S. medical students to negotiate uncertainty in clinical care: a CERA study. Fam Med. 2015;47(1):31-6.
38. Christensen-Salem A, Zanini MTF, Walumbwa FO, et al. Communal solidarity in extreme environments: The role of servant leadership and social resources in building serving culture and service performance. J Bus Res. 2021;135:829-39.
39. Cooke G, Tapley A, Holliday E, et al. Responses to clinical uncertainty in Australian general practice trainees: a cross-sectional analysis. Med Educ. 2017;51(12):1277-88.

Accepted Manuscript - BJGP Open - BJGPO.2023.0191

40. Schneider A, Bühner M, Herzog T, et al. Educational Intervention Reduced Family Medicine Residents' Intention to Request Diagnostic Tests: Results of a Controlled Trial. Med Decis Making. 2021;41(3):329-39.
41. DeForge BR, Sobal J. Intolerance of ambiguity among family practice residents. Fam Med. 1991;23(6):466-8.
42. Taylor D, Picker B, Woolever D, et al. A Pilot Study to Address Tolerance of Uncertainty Among Family Medicine Residents. Fam Med. 2018;50(7):531-8.
43. Guenter D, Fowler N, Lee L. Clinical uncertainty: helping our learners. Can Fam Physician. 2011;57(1):120-5.
44. Morgan S, Chan M, Starling C. Starting off in general practice - consultation skill tips for new GP registrars. Aust Fam Physician. 2014;43(9):645-8.
45. Elwyn G, Edwards A, Gwyn R, Grol R. Towards a feasible model for shared decision making: focus group study with general practice registrars. BMJ. 1999;319(7212):753-6.
46. Cooper M, Sornalingam S, Jegatheesan M, Fernandes C. The undergraduate 'corridor of uncertainty': teaching core concepts for managing clinical uncertainty as the 'special technique' of general practice. Educ Prim Care. 2022;33(2):120-4.
47. Carr D, Gormley GJ. 'Visiting uncertainty': an immersive primary care simulation to explore decision-making when there is clinical uncertainty. Educ Prim Care. 2022;33(4):237-43.
48. Parekh R, Jones MM, Singh S, et al. Medical students' experience of the hidden curriculum around primary care careers: a qualitative exploration of reflective diaries. BMJ Open. 2021;11(7):e049825.
49. Mamelok J. Is out-of-hours training for general practice registrars still relevant? General practice trainers' and general practice registrars' views on the provision of out-of-hours training. Educ Prim Care. 2005;16:150-9.
50. Tanna S, Fyfe M, Kumar S. Learning through service: a qualitative study of a community-based placement in general practice. Educ Prim Care. 2020;31(5):305-10.
51. Morgan S, Henderson KM, Tapley A, et al. Investigation of fatigue by Australian general practice registrars: a cross-sectional study. J Prim Health Care. 2015;7(2):109-16.
52. Howman M, Walters K, Rosenthal J, et al. "You kind of want to fix it don't you?" Exploring general practice trainees' experiences of managing patients with medically unexplained symptoms. BMC Med Educ. 2016;16(1):27.
53. Nevalainen M, Kuikka L, Sjoberg L, et al. Tolerance of uncertainty and fears of making mistakes among fifth-year medical students. Fam Med. 2012;44(4):240-6.
54. Griffiths J, Han, H., et al. . How family medicine residents learn. Understanding the role of cues in self-regulated learning. . Can Fam Physician. 2018;64(1).
55. Rossignac-Milon M, Higgins ET. Epistemic companions: shared reality development in close relationships. Curr Opin Psychol. 2018;23:66-71.
56. Ilgen JS EK, de Bruin A, et al. . Comfort with uncertainty: reframing our conceptions of how clinicians navigate complex clinical situations. Adv Health Sci Edu: Theory Pract. 2019;24(4):797-809.
57. Gardner NP, Gormley GJ, Kearney GP. Is there ever a single best answer (SBA): assessment driving certainty in the uncertain world of GP? Educ Prim Care. 2023:1-4.
58. WHO. Patient safety curriculum guide: multi-professional edition 2011 [Available:https://www.who.int/publications/i/item/9789241501958 [Accessed 24 November **2023**].
59. Eastwood JL, Koppelman-White E, Mi M, et al. Epistemic cognition in medical education: a literature review. Int J Med Educ. 2017;8:1-12.
60. Schwab A. Epistemic humility and medical practice: translating epistemic categories into ethical obligations. J Med Philos. 2012;37(1):28-48.

Accepted Manuscript. BJGP Open - BJGPO.2023.0191